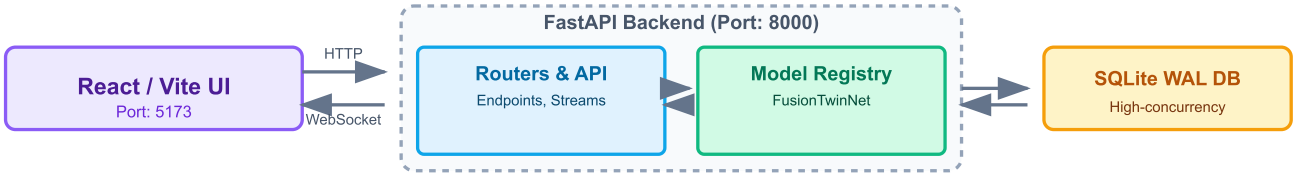
GlucoTwin System Architecture

This document outlines the high-level architecture of GlucoTwin, a probabilistic physiological digital twin platform designed for Type 2 Diabetes (T2D) management and real-time clinical decision support.



1. Presentation Layer (Frontend)

A modern Clinical Twin Dashboard built with React and Vite, served on port `5173`. It features real-time interactive charts built with Recharts.

- Consumes RESTful HTTP APIs for historic patient data and predictions.
- Maintains persistent WebSocket connections to visualize 15-minute sensor tick replays.

2. Application Layer (Backend API)

A high-performance FastAPI service running on port `8000`. It provides core routing logic and handles concurrent incoming requests via Uvicorn.

- `/predict`: Forecasts trajectories and alert probabilities.
- `/whatif`: Handles counterfactual simulations.
- `/stream`: Manages asynchronous telemetry streaming.

3. Machine Learning Core

The Model Registry manages a pre-warmed singleton instance of the PyTorch inference engine to ensure ultra-low latency.

- **FusionTwinNet:** Fuses static EHR features and dynamic sequence data.
- **Calibrator:** Applies isotonic regression for calibrated spike alerts.
- **TwinExplainer:** Computes physiological driver attributions.

4. Data Persistence Layer

A robust relational database optimized for high-concurrency read/write operations during real-time telemetry streaming.

- Utilizes SQLite configured in Write-Ahead Logging (WAL) mode.
- Stores anonymized profiles for 100 T2D patients.
- Houses over 576,000 historical CGM and vitals readings.